Assignment 5 — Optimization and Learning Rate Schedulers

Raisin dataset · 7–20–3–1 network · L1 λ = 0.7

Setup

900 observations; stratified 60/20/20 split; train-only standardization.
5,000 epochs, batch size 64, identical initialization and batch order.
Initial LR: 0.1 for GD/Momentum and 0.01 for Adam.

Final results

Optimizer	Scheduler	Val. loss	Val. accuracy
ADAM	constant	0.5515	85.56%
GD	Tutorial decay	0.5534	85.56%
ADAM	step	0.5582	86.11%
MOMENTUM	Tutorial decay	0.5588	86.67%
MOMENTUM	step	0.5640	86.67%
MOMENTUM	constant	0.5706	86.11%
GD	step	0.5731	86.67%
ADAM	Tutorial decay	0.5774	84.44%
GD	constant	0.5821	85.56%

Discussion

Tutorial decay lowered validation loss for GD (0.5821 → 0.5534) and Momentum (0.5706 → 0.5588), but raised it for Adam (0.5515 → 0.5774). Step decay gave GD and Momentum 86.67% validation accuracy. Adam with a constant rate had the lowest validation loss; its held-out test accuracy was 87.22%. The training/validation loss gap (about 0.26 vs 0.55–0.58) suggests overfitting. Accuracy and cross-entropy rank runs differently because confidence matters.

We selected by validation loss and used the test set once; a single split cannot establish a universal winner.

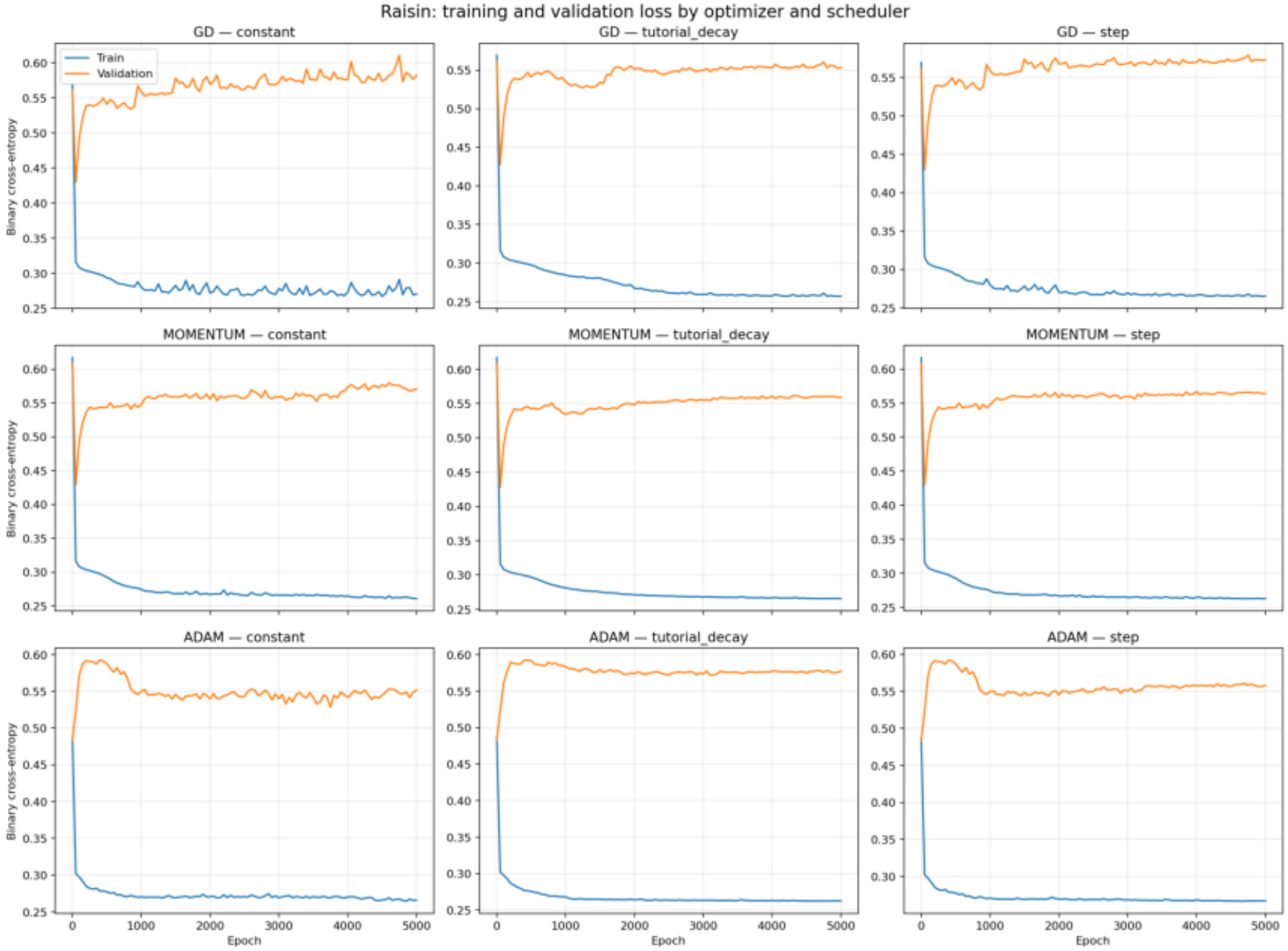
The tutorial's 'exponential' formula is inverse-time: $LR_0/(1 + 0.001 \times \text{epoch})$.

Step decay: $LR_0 \times 0.7^{(\text{floor}(\text{epoch} / 1000))}$. Selection used validation loss only.

Data: Çinar, Koklu & Tasdemir (2020), UCI Raisin, DOI: 10.24432/C5660T (CC BY 4.0).

Optimizer background: Kingma & Ba (2014), Adam. Course Optimization Methods notebook.

Training and validation loss



Learning rate schedules

